



# WORKSHOP

## (FITTING)

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## Introduction:

Machine tools are capable of producing work at a faster rate. but there are occasions when components are processed at the bench. Sometimes, it becomes necessary to replace or repair components which must be fit accurately with another component on reassembly. This involves a certain amount of hand fitting. The assembly of machine tools, jigs, gauges, etc, involves a certain amount of bench work. The accuracy of work done depends upon the experience and skill of the fitter. The term 'bench work' refers to the production of components by hand on the bench, whereas fitting deals with the assembly of mating parts, through removal of metal, to obtain the required fit. Both the bench work and fitting requires the use of a number of simple hand tools and considerable manual efforts. The operations in the above works consist of filing, chipping, scraping, sawing drilling and tapping.

## **Experiment: Make Internal and External threading on a square shaped Mild Steel**

**Aim:** To file a given Mild Steel (M.S) piece into a square shape of given dimension and to center punch the filed Mild Steel piece and make internal and external threading.

### **Tools Required:**

- Bench Vice
- Set of Files (Flat and Smooth)
- Steel Rule
- Try-Square
- Vernier Caliper
- Vernier Height Gauge
- Ball-Peen Hammer
- Scriber
- Dot Punch
- Angle Plate
- Surface Plate
- Centre Punch
- Drill Machine
- Drill bit
- Machine Vice
- Tap Set
- Tap Wrench
- Die Wrench
- Die

### **Operation:**

1. Measuring
2. Checking
3. Holding
4. Filing
5. Marking
6. Punching and Hammering
7. Drilling
8. Taping
9. Dyeing
10. Finishing

## Procedure:

1. Cutting the Mild Steel Strip:
  - The required size of the square mild steel strip is cut using a hacksaw from a longer steel strip.
2. Dimensions:
  - The dimensions of the cut piece are verified using a steel rule to ensure they are close to the desired size.
3. Filing Two Adjacent Sides:
  - The job is securely fixed in a bench vice.
  - The two adjacent sides are filed using a rough flat file first, followed by a smooth flat file, ensuring that the two sides are at a right angle to each other.
  - The right angle between the two adjacent sides is checked using a try-square.
4. Marking the Workpiece:
  - Apply chalk to the surface of the workpiece for better visibility.
  - Mark the dimensions by scribing two lines, using the Vernier height gauge, with reference to the two datum sides. Utilize an angle plate and surface plate for accuracy.
5. Filing the Remaining Sides:
  - The workpiece is refitted in the bench vice, and the remaining two sides are filed. Check the flatness of the surfaces after filing. As filing removes material more precisely and gradually than sawing, it is preferred for this final shaping.

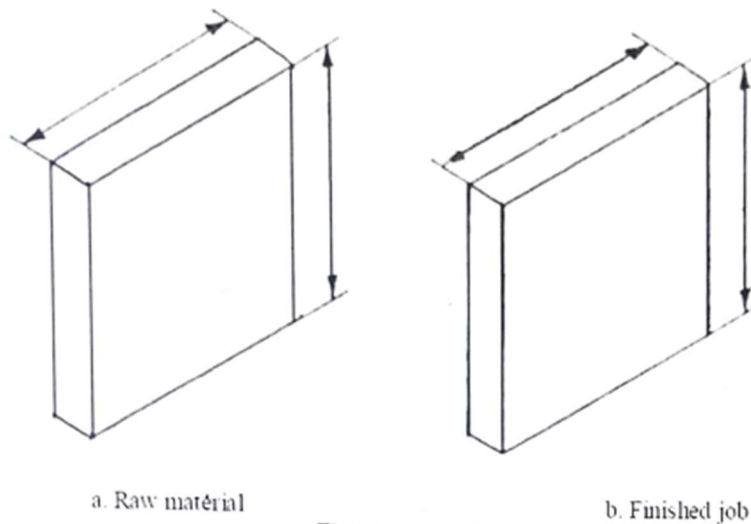


Figure: Square filing

6. Marking the Centre:

- After filing, place the square workpiece on a flat surface. Use a steel rule and vernier height gauge to find the center of the workpiece.

-Scribe a line along the center using a scribe to mark the position. Cross-check the center mark with a try-square for accuracy.

7. Punching the Centre Mark:

- Place the workpiece back in the bench vice, ensuring it is stable. Position the dot punch precisely on the scribed center mark. Lightly tap the dot punch with a ball-peen hammer to create a small indentation. Follow up with a center punch placed over the dot punch mark and strike it firmly with the hammer to deepen the indentation. This ensures accurate drilling.

8. Drilling the Hole:

- Secure the workpiece on a drilling machine table using clamps. Align the drill bit with the center-punched mark. Start the drilling machine and carefully drill a hole of the required diameter. Ensure that the hole depth is appropriate for the length of the cylindrical rod that will be threaded and inserted.

9. Internal Threading:

- After drilling, the hole is prepared for internal threading. Select the correct tap size corresponding to the hole diameter. Attach the tap to a tap wrench and apply cutting fluid to reduce friction. Insert the tap into the hole and rotate it clockwise to cut the threads inside the hole. After completing the threading, reverse the tap to remove it, clearing any debris.

10. External Threading:

- Secure the cylindrical rod in the bench vice. Select the appropriate die corresponding to the external diameter of the rod. Place the die in a die holder and apply cutting fluid to the rod. Position the die at the end of the rod and rotate the die holder clockwise to start cutting the external threads. Continue threading until the desired length is fully threaded. Reverse the die holder to remove it, ensuring clean, well-formed thread.

11. Fitting the Threaded Rod:

- Check the alignment and fit of the threaded rod in the hole. Ensure that the threads are clean and free of any burrs or imperfections.

